

25CS1302E - DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT

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SECTION: 2A

LAB COURSEWORK

Q. Week 2:

The Librarian's Dashboard Scenario The library is now digital! The head librarian needs a report to see which members are active and which books are in the collection. You also need to handle a "Book Donation" process where a new book is added and assigned to a category at the exact same time—if one part fails, the whole process must stop to keep the catalog accurate.

Task to be Done

1. The Catalog Search (JOINS): Create a third table called Loans that connects member_id to book_id. Write a query to show the Member Name and the Book Title they have currently borrowed.
 2. Collection Stats (Aggregations): Write a query to find the total number of books published in each year. o
- Hint: Use COUNT(book_id) and GROUP BY published_year.
3. The Donation Transaction (ACID): Use a Transaction (BEGIN...COMMIT) to add a new book to the Books table and simultaneously log the entry in a Donation_History table.
 4. Search Speed (Indexing): Create an Index on the isbn column. Explain why searching by ISBN is now

faster for the librarian than searching by Title. Skill Set Gained • Relational Logic:

Connecting independent

tables (Books + Members) through a Junction table (Loans). • Analytical SQL: Using GROUP BY to generate

business insights. • Atomic Operations: Using Transactions to ensure multiple database changes succeed or

fail as a single unit.

SQL: mysql> USE bookflow_db;

Database changed

mysql>

mysql> SHOW TABLES;

```
+-----+
| Tables_in_bookflow_db |
+-----+
| books                  |
| members                |
+-----+
```

2 rows in set (0.005 sec)

mysql> SELECT * FROM books;

```
+-----+-----+-----+-----+
| book_id | title      | isbn      | published_year |
+-----+-----+-----+-----+
| 1       | The Alchemist | 9780061122415 | 1988 |
| 2       | Clean Code   | 9780132350884 | 2008 |
| 3       | Atomic Habits | 9780735211292 | 2018 |
+-----+-----+-----+-----+
```

3 rows in set (0.001 sec)

mysql> SELECT * FROM members;

```
+-----+-----+-----+
| member_id | full_name | email                |
+-----+-----+-----+
| 1         | Anil Kumar | anil.kumar@example.com |
| 2         | Priya Sharma | priya.sharma@example.com |
| 3         | Ravi Verma | ravi.verma@example.com |
+-----+-----+-----+
```

3 rows in set (0.000 sec)

mysql> CREATE TABLE Loans (

```
-> loan_id INT PRIMARY KEY,
-> member_id INT,
-> book_id INT,
-> loan_date DATE,
-> FOREIGN KEY (member_id) REFERENCES members(member_id),
-> FOREIGN KEY (book_id) REFERENCES books(book_id)
-> );
```

Query OK, 0 rows affected (0.027 sec)

mysql> SHOW TABLES;

```
+-----+
| Tables_in_bookflow_db |
+-----+
| books                  |
| Loans                  |
+-----+
```

| members |

+-----+

3 rows in set (0.001 sec)

mysql> DESCRIBE Loans;

+-----+-----+-----+-----+-----+-----+

| Field | Type | Null | Key | Default | Extra |

+-----+-----+-----+-----+-----+-----+

| loan_id | int | NO | PRI | NULL | |

| member_id | int | YES | MUL | NULL | |

| book_id | int | YES | MUL | NULL | |

| loan_date | date | YES | | NULL | |

+-----+-----+-----+-----+-----+-----+

4 rows in set (0.006 sec)

mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date)

-> VALUES

-> (1, 1, 1, '2025-01-05'),

-> (2, 2, 2, '2025-01-08'),

-> (3, 3, 3, '2025-01-10'),

-> (4, 1, 2, '2025-02-01'),

-> (5, 2, 1, '2025-02-05'),

-> (6, 3, 2, '2025-02-12'),

-> (7, 1, 3, '2025-03-01'),

-> (8, 2, 3, '2025-03-07'),

-> (9, 3, 1, '2025-03-15'),

-> (10, 1, 1, '2025-04-01');

Query OK, 10 rows affected (0.007 sec)

Records: 10 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Loans;

+-----+-----+-----+-----+

| loan_id | member_id | book_id | loan_date |

+-----+-----+-----+-----+

| 1 | 1 | 1 | 2025-01-05 |

| 2 | 2 | 2 | 2025-01-08 |

| 3 | 3 | 3 | 2025-01-10 |

| 4 | 1 | 2 | 2025-02-01 |

| 5 | 2 | 1 | 2025-02-05 |

| 6 | 3 | 2 | 2025-02-12 |

| 7 | 1 | 3 | 2025-03-01 |

| 8 | 2 | 3 | 2025-03-07 |

| 9 | 3 | 1 | 2025-03-15 |

| 10 | 1 | 1 | 2025-04-01 |

+-----+-----+-----+-----+

10 rows in set (0.001 sec)

mysql> SELECT m.full_name AS Member_Name,

```

-> b.title AS Book_Title
-> FROM Loans l
-> INNER JOIN members m
-> ON l.member_id = m.member_id
-> INNER JOIN books b
-> ON l.book_id = b.book_id;

```

```

+-----+-----+
| Member_Name | Book_Title |
+-----+-----+
| Ravi Verma   | The Alchemist |
| Priya Sharma | The Alchemist |
| Anil Kumar   | The Alchemist |
| Anil Kumar   | The Alchemist |
| Ravi Verma   | Clean Code    |
| Priya Sharma | Clean Code    |
| Anil Kumar   | Clean Code    |
| Ravi Verma   | Atomic Habits |
| Priya Sharma | Atomic Habits |
| Anil Kumar   | Atomic Habits |
+-----+-----+

```

10 rows in set (0.001 sec)

```

mysql> SELECT published_year,
-> COUNT(book_id) AS Total_Books
-> FROM books
-> GROUP BY published_year
-> ORDER BY published_year;

```

```

+-----+-----+
| published_year | Total_Books |
+-----+-----+
| 1988          | 1          |
| 2008          | 1          |
| 2018          | 1          |
+-----+-----+

```

3 rows in set (0.003 sec)

```

mysql> CREATE TABLE Donation_History (
-> donation_id INT PRIMARY KEY,
-> book_id INT,
-> donor_name VARCHAR(100),
-> donation_date DATE,
-> FOREIGN KEY (book_id) REFERENCES books(book_id)
-> );

```

Query OK, 0 rows affected (0.011 sec)

```
mysql> START TRANSACTION;
```

Query OK, 0 rows affected (0.000 sec)

```
mysql> INSERT INTO books (title, isbn, published_year)
-> VALUES ('Animal Farm', '9780451526342', 1945);
```

Query OK, 1 row affected (0.003 sec)

```
mysql> SELECT book_id
-> FROM books
-> WHERE isbn = '9780451526342';
```

```
+-----+
| book_id |
+-----+
|    5 |
+-----+
```

1 row in set (0.001 sec)

```
mysql> INSERT INTO Donation_History
-> (donation_id, book_id, donor_name, donation_date)
-> VALUES
-> (1, 5, 'Raj Kumar', CURDATE());
```

Query OK, 1 row affected (0.004 sec)

```
mysql> COMMIT;
```

Query OK, 0 rows affected (0.002 sec)

```
mysql> SELECT * FROM books;
```

```
+-----+-----+-----+-----+
| book_id | title      | isbn      | published_year |
+-----+-----+-----+-----+
| 1 | The Alchemist | 9780061122415 | 1988 |
| 2 | Clean Code   | 9780132350884 | 2008 |
| 3 | Atomic Habits | 9780735211292 | 2018 |
| 5 | Animal Farm  | 9780451526342 | 1945 |
+-----+-----+-----+-----+
```

4 rows in set (0.000 sec)

```
mysql> SELECT * FROM Donation_History;
```

```
+-----+-----+-----+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+-----+-----+-----+
| 1 | 5 | Raj Kumar | 2026-07-31 |
+-----+-----+-----+-----+
```

1 row in set (0.000 sec)

```
mysql> START TRANSACTION;
```

Query OK, 0 rows affected (0.000 sec)

```
mysql> INSERT INTO books (title, isbn, published_year)
-> VALUES ('Temporary Book', '111111111111', 2025);
```

Query OK, 1 row affected (0.001 sec)

```
mysql> SELECT * FROM books
-> WHERE title = 'Temporary Book';
```

```
+-----+-----+-----+-----+
```

book_id	title	isbn	published_year
6	Temporary Book	1111111111111	2025

1 row in set (0.000 sec)

mysql> ROLLBACK;

Query OK, 0 rows affected (0.001 sec)

mysql> SELECT * FROM books

-> WHERE title = 'Temporary Book';

Empty set (0.000 sec)

mysql> CREATE INDEX idx_title

-> ON books(title);

Query OK, 0 rows affected (0.029 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> SHOW INDEX FROM books;

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
books	0	PRIMARY	1	book_id	A	5			NULL	BTREE				
books	0		1	isbn	A	5			NULL	BTREE				
books	1	idx_title	1	title	A	4			NULL	BTREE				

3 rows in set (0.005 sec)

mysql>